TEST 2 MEMO CRC CARD REVISION WORKSHEET

Class / Interface - Responsibility - Collaborator - Design Evolution

Project: Test 2 Memo **Team:** Group 7

DIRECTIONS

Start with the CRC cards submitted for Test 1 - do not redesign the system from scratch.
Carry forward everything that is still accurate. Revise responsibilities and collaborators when the Test 2 use cases reveal a needed change, and add a class or interface only when it is justified by the current design. Mark each card as **Revised from Test 1** or **New for Test 2**, explain the change if revised, and keep the CRC cards, UML diagram, and C# implementation consistent. Duplicate this page if more cards are needed.

CRC CARD 1

Class / Interface: `Bracket` **Status:** [X] Revised from Test 1 [ ] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Keeping overall structure of the matches	1. Match
2. Update with each outcome	2. IBracketType
3. Categorizing matches into Winner or Loser brackets	3. _____
4. _____	4. _____

Test 2 change / justification: Made bracket type follow abstraction instead of a string

CRC CARD 2

Class / Interface: Team **Status:** [X] Revised from Test 1 [ ] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Encapsulates core team identity (name and logo)	1. Player
2. Tracks individual team performance metrics (wins, losses, and elimination status)	2. _____
3. Manages the internal player roster	3. _____
4. Updates mutable team properties through encapsulated behavior methods	4. _____

Test 2 change / justification: Added additional functionality with players

CRC CARD 3

Class / Interface: Match **Status:** [X] Revised from Test 1 [ ] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Show which team versus each other	1. Team
2. Schedule the match and show what time	2. _____
3. Update team records upon match completion	3. _____
4. _____	4. _____

Test 2 change / justification: Added winner and round number (in terms of matches not actual game) attributes

DESIGN EVOLUTION CHECK

[ ] Started from the submitted Test 1 CRC cards
[ ] Carried forward all still-accurate design content
[ ] Each revision traces to a Test 2 use case
[ ] Each new class/interface has a clear justification

[ ] Responsibilities/collaborators match UML and C#
[ ] Service dependencies use appropriate abstractions
[ ] Domain objects still protect their invariants
[ ] Current design replaces inaccurate Test 1 details

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CRC CARD 4

Class / Interface: `TeamService` **Status:** [X] Revised from Test 1 [ ] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Manages the internal collection and data storage of teams	1. `Team`
2. _____	2. _____
3. _____	3. _____
4. _____	4. _____

Test 2 change / justification: Changed method name to be more clear (UpdateTeam -> UpdateTeamName)

CRC CARD 5

Class / Interface: Player **Status:** [ ] Revised from Test 1 [X] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Represents a single team member entity within a team roster.	1. `PlayerStats`
2. Encapsulates individual participant information (name and email)	2. _____
3. _____	3. _____
4. _____	4. _____

Test 2 change / justification: Expands the Team domain model to support detailed roster tracking and rich domain behavior.

CRC CARD 6

Class / Interface: PlayerService **Status:** [ ] Revised from Test 1 [X] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Manages the collection and persistence lifecycle of individual players	1. `Player`
2. Provides lookup operations for player profiles by email.	2. _____
3. Exposes a safe, read-only collection of all registered players.	3. _____
4. _____	4. _____

Test 2 change / justification: Ensures individual players can be registered, retrieved, and managed independently from teams

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CRC CARD 7

Class / Interface: `PlayerStats` **Status:** [ ] Revised from Test 1 [X] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Defines the polymorphic contract for game-specific stats.	1. `Player`
2. Encapsulates individual participant information (name and email)	2. _____
3. _____	3. _____
4. _____	4. _____

Test 2 change / justification: Added attribute to player to represent different games

CRC CARD 8

Class / Interface: IBracketType Status: [ ] Revised from Test 1 [X] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Expose a readable name representing the tournament format	1. `Team`
2. Define a uniform contract for tournament formatting and matching algorithms	2. Match
3. Abstractly generate structural match lists from a collection of participating teams.	3. _____
4. _____	4. _____

Test 2 change / justification: More extensible design for different tournament types (e.g, single elimination, double elimination)

CRC CARD 9

Class / Interface: MatchSchedulingService Status: [ ] Revised from Test 1 [X] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Generates tournament brackets based on selected tournament type.	1. `Bracket`
2. Initializes and manages the collection of scheduled matches	2. Match
3. Handles match rescheduling and time adjustments	3. Team
4. _____	4. IBracketType

Test 2 change / justification: Centralizes bracket generation and scheduling logic so the application can dynamically structure tournament formats

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CRC CARD 10

Class / Interface: `TournamentProgressionService` **Status:** [ ] Revised from Test 1 [X] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Tracks active brackets currently underway in the system	1. `Bracket`
2. Records match results and updates corresponding participant records	2. Match
3. Advances winning competitors to subsequent slots in the bracket tree	3. Team
4. _____	4. IMatchSchedulingService 5. ITeamService

Test 2 change / justification: Manages the flow and state transitions of an ongoing tournament so match outcomes correctly advances through future rounds.

CRC CARD 11

Class / Interface: LeaderboardService **Status:** [ ] Revised from Test 1 [X] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES
1. Dynamically computes and sorts team standings based on wins and performance stats.	1. `Team`
2. Generates player rankings based on their stats	2. Player
3. _____	3. ITeamService
4. _____	4. IPlayerService

Test 2 change / justification: More extensible design for different tournament types (e.g, single elimination, double elimination)

CRC CARD 12

Class / Interface: **Status:** [ ] Revised from Test 1 [ ] New for Test 2

RESPONSIBILITIES	COLLABORATORS / DEPENDENCIES

Test 2 change / justification: Centralizes match outcome tracking, elimination state updates, and loser bracket routing.

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